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Chapter 32 - Classification And Characteristic Coordinates

Chapter 31 introduced the main PDE models and used separation of variables in fixed-boundary heat and wave problems. This chapter looks more closely at the second-order part of a PDE.

The signs and combinations of the second derivatives tell us which coordinate directions are natural for the equation. For a hyperbolic equation, those directions lead to characteristic curves. For parabolic and elliptic equations, related coordinate changes reveal different canonical forms.

This chapter will:

- isolate the principal part of a second-order PDE
- classify an equation as hyperbolic, parabolic, or elliptic
- explain why classification may change from one point to another
- derive the characteristic equation from a level curve
- distinguish a characteristic curve from its slope
- transform first and second partial derivatives carefully
- check that a coordinate change is locally invertible
- reduce constant-coefficient equations to canonical form
- solve simple canonical hyperbolic and parabolic equations
- use dimensional consistency to check PDE models
- nondimensionalise heat, wave, and advection-diffusion equations
- build a reliable classification-and-transformation workflow

Reading The Principal Part

The Goal Of This Block

The goal is to identify the part of a second-order PDE that controls its type and to apply the discriminant test without losing the factor of 2 in the mixed term.

classification begins with the highest-order derivatives, not with every term in the equation.

The General Second-Order Form

For an unknown function $u = u(x, y)$, write a general second-order equation as:

$$A u_{xx} + 2B u_{xy} + C u_{yy} + D u_x + E u_y + F u = G.$$

The coefficients may depend on x and y . For example:

$$A = A(x, y), \quad B = B(x, y), \quad C = C(x, y).$$

The **principal part** is the collection of highest-order terms:

$$\boxed{A u_{xx} + 2B u_{xy} + C u_{yy}}.$$

The lower-order terms Du_x , Eu_y , and Fu affect the solutions, but they do not determine the second-order type.

The Factor-Of-Two Convention

This chapter uses:

$$A u_{xx} + 2B u_{xy} + C u_{yy}.$$

Therefore, if the displayed equation contains:

$$6u_{xy},$$

then:

$$2B = 6,$$

so:

$$B = 3.$$

Do not insert the visible coefficient 6 directly into $B^2 - AC$.

The practical rule is:

first match the mixed term with $2Bu_{xy}$; only then calculate the discriminant.

The Discriminant Test

Define:

$$\Delta = B^2 - AC.$$

For $\Delta = B^2 - AC$, the classification at the point being considered is:

$$\begin{array}{l|l} \Delta > 0 & \text{hyperbolic} \\ \Delta = 0 & \text{parabolic} \\ \Delta < 0 & \text{elliptic} \end{array}$$

The same information is contained in the symmetric principal matrix:

$$M = \begin{pmatrix} A & B \\ B & C \end{pmatrix}.$$

Its determinant is:

$$\begin{aligned} \det M &= AC - B^2 \\ &= -\Delta. \end{aligned}$$

Thus the PDE discriminant has the opposite sign from the determinant of the principal matrix.¹

Worked Classification With A Mixed Term

Classify:

$$3u_{xx} - 8u_{xy} + 2u_{yy} + u_x = 0.$$

Match the principal part with:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy}.$$

¹ The matrix packages the quadratic expression $Ap^2 + 2Bpq + Cq^2$. Hyperbolic, parabolic, and elliptic behaviour correspond to that quadratic expression having two, one, or no real zero directions.

For $3u_{xx} - 8u_{xy} + 2u_{yy} + u_x = 0$, the principal coefficients are:

$$A = 3, \quad 2B = -8, \quad C = 2.$$

Solve $2B = -8$ for B :

$$B = -4.$$

Now calculate the discriminant:

$$\begin{aligned} \Delta &= B^2 - AC \\ &= (-4)^2 - (3)(2) \\ &= 16 - 6 \\ &= 10. \end{aligned}$$

Since $\Delta > 0$:

the equation is hyperbolic.

The first-order term u_x does not enter this classification.

Worked Classification At Different Points

Consider:

$$(1 - x^2 - y^2)u_{xx} + u_{yy} = 0.$$

Here:

$$A = 1 - x^2 - y^2, \quad B = 0, \quad C = 1.$$

For this variable-coefficient equation, the discriminant is:

$$\begin{aligned}\Delta &= B^2 - AC \\ &= 0 - (1 - x^2 - y^2)(1) \\ &= x^2 + y^2 - 1.\end{aligned}$$

Inside the unit circle, $x^2 + y^2 < 1$, so:

$$\Delta < 0.$$

The equation is elliptic there.

On the unit circle, $x^2 + y^2 = 1$, so:

$$\Delta = 0.$$

The equation is parabolic there.

Outside the unit circle, $x^2 + y^2 > 1$, so:

$$\Delta > 0.$$

The equation is hyperbolic there.

What this tells us:

**a variable-coefficient PDE can change type across a curve.
Classification is then a local statement, not one permanent label for the whole plane.**

Block 1 Summary

For:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + \text{lower-order terms} = 0,$$

calculate:

$$\Delta = B^2 - AC.$$

The sign of Δ determines the type. If the coefficients depend on the independent variables, repeat the sign analysis in each relevant region.

Changing Coordinates Without Skipping The Chain Rule

The Goal Of This Block

The goal is to express a PDE in new coordinates while keeping every chain-rule term visible. These formulas explain exactly how characteristic and canonical coordinates change the principal part.

Old And New Variables

Introduce new coordinates:

$$\xi = \xi(x, y), \quad \eta = \eta(x, y).$$

Write the same dependent variable in the new coordinates as:

$$u(x, y) = U(\xi, \eta).$$

The letters u and U do not represent two unrelated solutions. They are two coordinate descriptions of the same function.

The coordinate change is locally invertible when its Jacobian is nonzero:

$$J = \frac{\partial(\xi, \eta)}{\partial(x, y)} = \xi_x \eta_y - \xi_y \eta_x \neq 0.$$

If $J = 0$, the two proposed coordinates fail to provide two independent directions.

Transform The First Derivatives

Since $u(x, y) = U(\xi(x, y), \eta(x, y))$, the multivariable chain rule gives:

$$u_x = U_\xi \xi_x + U_\eta \eta_x.$$

Similarly:

$$u_y = U_\xi \xi_y + U_\eta \eta_y.$$

Each old derivative differentiates through both new coordinates. For example, u_x contains one contribution from the change of ξ with x and another from the change of η with x .

Derive The Second X-Derivative

Start from the transformed first derivative:

$$u_x = U_\xi \xi_x + U_\eta \eta_x.$$

Differentiate both terms with respect to x :

$$u_{xx} = \frac{\partial}{\partial x}(U_\xi \xi_x) + \frac{\partial}{\partial x}(U_\eta \eta_x).$$

Apply the product rule to both terms in the expression for u_{xx} :

$$u_{xx} = (U_{\xi})_x \tilde{\zeta}_x + U_{\xi} \tilde{\zeta}_{xx} + (U_{\eta})_x \eta_x + U_{\eta} \eta_{xx}.$$

Now apply the chain rule to the derivatives of U_{ξ} and U_{η} :

$$(U_{\xi})_x = U_{\xi\xi} \tilde{\zeta}_x + U_{\xi\eta} \eta_x,$$

$$(U_{\eta})_x = U_{\eta\xi} \tilde{\zeta}_x + U_{\eta\eta} \eta_x.$$

Assuming the relevant mixed derivatives are continuous, $U_{\eta\xi} = U_{\xi\eta}$. Substitute the two chain-rule expressions into u_{xx} :

$$\begin{aligned} u_{xx} = & U_{\xi\xi} \tilde{\zeta}_x^2 + 2U_{\xi\eta} \tilde{\zeta}_x \eta_x + U_{\eta\eta} \eta_x^2 \\ & + U_{\xi} \tilde{\zeta}_{xx} + U_{\eta} \eta_{xx}. \end{aligned}$$

The first line contains the new second derivatives. The second line contributes only lower-order derivatives of U .

Transform The Mixed And Y-Derivatives

Applying the same chain rule to u_x with respect to y gives:

$$\begin{aligned} u_{xy} = & U_{\xi\xi} \tilde{\zeta}_x \tilde{\zeta}_y + U_{\xi\eta} (\tilde{\zeta}_x \eta_y + \tilde{\zeta}_y \eta_x) + U_{\eta\eta} \eta_x \eta_y \\ & + U_{\xi} \tilde{\zeta}_{xy} + U_{\eta} \eta_{xy}. \end{aligned}$$

Differentiating u_y once more with respect to y gives:

$$\begin{aligned} u_{yy} = & U_{\xi\xi} \tilde{\zeta}_y^2 + 2U_{\xi\eta} \tilde{\zeta}_y \eta_y + U_{\eta\eta} \eta_y^2 \\ & + U_{\xi} \tilde{\zeta}_{yy} + U_{\eta} \eta_{yy}. \end{aligned}$$

For a linear coordinate change, all second derivatives of ξ and η are zero. The lower-order terms in these boxes then disappear.

Read The New Principal Coefficients

Substitute the transformed derivatives into:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy}.$$

The coefficient of $U_{\xi\xi}$ is:

$$\bar{A} = A\tilde{\xi}_x^2 + 2B\tilde{\xi}_x\tilde{\xi}_y + C\tilde{\xi}_y^2.$$

The coefficient of $U_{\eta\eta}$ is:

$$\bar{C} = A\eta_x^2 + 2B\eta_x\eta_y + C\eta_y^2.$$

The transformed equation uses $2\bar{B}U_{\xi\eta}$. Its coefficient is:

$$\begin{aligned} 2\bar{B} = 2[A\tilde{\xi}_x\eta_x + B(\tilde{\xi}_x\eta_y + \tilde{\xi}_y\eta_x) \\ + C\tilde{\xi}_y\eta_y]. \end{aligned}$$

The remaining chain-rule pieces multiply U_{ξ} and U_{η} , so they become lower-order terms in the transformed PDE.

A Linear Coordinate Check

Let:

$$\tilde{\xi} = x + y, \quad \eta = 2x - y.$$

The first derivatives are:

$$\begin{aligned}\tilde{\zeta}_x &= 1, & \tilde{\zeta}_y &= 1, \\ \eta_x &= 2, & \eta_y &= -1.\end{aligned}$$

Calculate the Jacobian:

$$\begin{aligned}J &= \tilde{\zeta}_x \eta_y - \tilde{\zeta}_y \eta_x \\ &= (1)(-1) - (1)(2) \\ &= -3.\end{aligned}$$

Since $J \neq 0$, the coordinates are independent.

All second derivatives of $\tilde{\zeta}$ and η vanish because both coordinates are linear functions of x and y . This is why linear changes are especially convenient for constant-coefficient PDEs.

Block 2 Summary

A valid coordinate change needs $J \neq 0$. The chain rule transforms each old second derivative into a combination of $U_{\tilde{\zeta}\tilde{\zeta}}$, $U_{\tilde{\zeta}\eta}$, $U_{\eta\eta}$, and possibly lower-order terms. Canonical coordinates are chosen to make one or more principal coefficients vanish.

Deriving Characteristic Curves

The Goal Of This Block

The goal is to derive the characteristic equation and connect its real roots to the three PDE types.

Choose A Coordinate That Removes One Pure Derivative

Suppose one new coordinate is:

$$\xi = \phi(x, y).$$

The coefficient of $U_{\xi\xi}$ in the transformed principal part is:

$$\bar{A} = A\phi_x^2 + 2B\phi_x\phi_y + C\phi_y^2.$$

To remove $U_{\xi\xi}$, require:

$$\boxed{A\phi_x^2 + 2B\phi_x\phi_y + C\phi_y^2 = 0}.$$

The curves $\phi(x, y) = \text{constant}$ that satisfy this condition are characteristic curves.

Convert The Level-Curve Condition Into A Slope Equation

Along a level curve $\phi(x, y) = k$, the differential of ϕ is zero:

$$d\phi = 0.$$

Expand the level-curve condition $d\phi = 0$:

$$\phi_x dx + \phi_y dy = 0.$$

If $dx \neq 0$, define the curve slope:

$$m = \frac{dy}{dx}.$$

Divide the level-curve equation by dx :

$$\phi_x + \phi_y m = 0.$$

Solve for ϕ_x :

$$\phi_x = -m\phi_y.$$

Substitute this into the characteristic condition:

$$A\phi_x^2 + 2B\phi_x\phi_y + C\phi_y^2 = 0.$$

This gives:

$$A(-m\phi_y)^2 + 2B(-m\phi_y)\phi_y + C\phi_y^2 = 0.$$

Expand the powers and products:

$$Am^2\phi_y^2 - 2Bm\phi_y^2 + C\phi_y^2 = 0.$$

For a nondegenerate level curve with $\phi_y \neq 0$, divide by ϕ_y^2 :

$$\boxed{Am^2 - 2Bm + C = 0}.$$

Equivalently:

$$\boxed{A(dy)^2 - 2B dx dy + C(dx)^2 = 0}.$$

The differential form remains useful when a characteristic is vertical and dy/dx is not finite.

Why The Discriminant Reappears

For $A \neq 0$, solve:

$$Am^2 - 2Bm + C = 0$$

with the quadratic formula:

$$\begin{aligned} m &= \frac{2B \pm \sqrt{(-2B)^2 - 4AC}}{2A} \\ &= \frac{2B \pm \sqrt{4B^2 - 4AC}}{2A} \\ &= \frac{2B \pm 2\sqrt{B^2 - AC}}{2A} \\ &= \frac{B \pm \sqrt{B^2 - AC}}{A}. \end{aligned}$$

Since $\Delta = B^2 - AC$:

$$\boxed{m = \frac{B \pm \sqrt{\Delta}}{A}}.$$

Therefore:

- $\Delta > 0$ gives two distinct real characteristic directions
- $\Delta = 0$ gives one repeated real characteristic direction
- $\Delta < 0$ gives no real characteristic direction

This is the geometric reason behind the hyperbolic, parabolic, and elliptic classification.

Characteristic Direction Versus Characteristic Curve

The formula:

$$\frac{dy}{dx} = m(x, y)$$

specifies a direction field. A characteristic curve is obtained by solving that first-order ODE.

If m is constant, integration gives straight lines. For example:

$$\frac{dy}{dx} = -2$$

integrates to:

$$y = -2x + k.$$

Rearrange to place the constant on one side:

$$2x + y = k.$$

Thus a suitable characteristic coordinate is:

$$\xi = 2x + y.$$

Do not stop after finding the slope. The coordinate used in the transformation must be constant along the corresponding family of curves.

A Quick Characteristic Calculation

Find the characteristic slopes of:

$$2u_{xx} + 2u_{xy} - u_{yy} = 0.$$

Match the principal coefficients:

$$A = 2, \quad 2B = 2, \quad C = -1.$$

Hence:

$$B = 1.$$

The slope equation is:

$$Am^2 - 2Bm + C = 0.$$

Substitute $A = 2$, $B = 1$, and $C = -1$:

$$2m^2 - 2m - 1 = 0.$$

Apply the quadratic formula:

$$\begin{aligned} m &= \frac{2 \pm \sqrt{(-2)^2 - 4(2)(-1)}}{2(2)} \\ &= \frac{2 \pm \sqrt{4 + 8}}{4} \\ &= \frac{2 \pm 2\sqrt{3}}{4} \\ &= \frac{1 \pm \sqrt{3}}{2}. \end{aligned}$$

The two real slopes confirm that the equation is hyperbolic.

Block 3 Summary

Characteristic coordinates satisfy:

$$A\phi_x^2 + 2B\phi_x\phi_y + C\phi_y^2 = 0.$$

For nonvertical curves, their slopes satisfy:

$$A\left(\frac{dy}{dx}\right)^2 - 2B\frac{dy}{dx} + C = 0.$$

Solve the slope equation, integrate each direction field, and use the resulting constants of integration as new coordinates.

A Hyperbolic Equation In Canonical Coordinates

The Goal Of This Block

The goal is to find both characteristic families, build a valid coordinate change, and reduce a hyperbolic PDE to a mixed-derivative canonical form.

Classify And Find The Characteristic Slopes

Consider:

$$u_{xx} - 4u_{xy} + 3u_{yy} = 0.$$

Match the equation with:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} = 0.$$

For $u_{xx} - 4u_{xy} + 3u_{yy} = 0$, the principal coefficients are:

$$A = 1, \quad 2B = -4, \quad C = 3,$$

so:

$$B = -2.$$

Calculate the discriminant:

$$\begin{aligned}\Delta &= B^2 - AC \\ &= (-2)^2 - (1)(3) \\ &= 1 > 0.\end{aligned}$$

The equation is hyperbolic. Its characteristic slope equation is:

$$Am^2 - 2Bm + C = 0.$$

Insert $A = 1$, $B = -2$, and $C = 3$:

$$m^2 + 4m + 3 = 0.$$

Factor:

$$(m + 1)(m + 3) = 0.$$

Therefore the two slopes are:

$$m_1 = -1, \quad m_2 = -3.$$

Integrate Both Characteristic Families

For the first family:

$$\frac{dy}{dx} = -1.$$

Integrate with respect to x :

$$y = -x + k_1.$$

Rearrange $y = -x + k_1$:

$$x + y = k_1.$$

A coordinate that is constant on this first family is:

$$\xi = x + y.$$

For the second family:

$$\frac{dy}{dx} = -3.$$

Integrate:

$$y = -3x + k_2.$$

Rearrange $y = -3x + k_2$:

$$3x + y = k_2.$$

A coordinate that is constant on this second family is:

$$\eta = 3x + y.$$

These coordinates are constant along the two characteristic families.

Check The Jacobian

For:

$$\xi = x + y, \quad \eta = 3x + y,$$

the coordinate derivatives are:

$$\begin{aligned} \xi_x &= 1, & \xi_y &= 1, \\ \eta_x &= 3, & \eta_y &= 1. \end{aligned}$$

Calculate:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (1)(1) - (1)(3) \\ &= -2. \end{aligned}$$

Since $J \neq 0$, the transformation is locally invertible.

Transform The Derivatives Explicitly

Write:

$$u(x, y) = U(\xi, \eta).$$

Because ξ and η are linear, their second derivatives vanish. The first derivative with respect to x is:

$$\begin{aligned} u_x &= U_\xi \xi_x + U_\eta \eta_x \\ &= U_\xi + 3U_\eta. \end{aligned}$$

Differentiate this expression with respect to x :

$$\begin{aligned} u_{xx} &= (U_\xi)_x + 3(U_\eta)_x \\ &= (U_{\xi\xi} + 3U_{\xi\eta}) + 3(U_{\eta\xi} + 3U_{\eta\eta}) \\ &= U_{\xi\xi} + 6U_{\xi\eta} + 9U_{\eta\eta}. \end{aligned}$$

The first derivative with respect to y is:

$$\begin{aligned} u_y &= U_\xi \xi_y + U_\eta \eta_y \\ &= U_\xi + U_\eta. \end{aligned}$$

Differentiate $u_x = U_\xi + 3U_\eta$ with respect to y :

$$\begin{aligned} u_{xy} &= (U_\xi)_y + 3(U_\eta)_y \\ &= (U_{\xi\xi} + U_{\xi\eta}) + 3(U_{\eta\xi} + U_{\eta\eta}) \\ &= U_{\xi\xi} + 4U_{\xi\eta} + 3U_{\eta\eta}. \end{aligned}$$

Differentiate $u_y = U_\xi + U_\eta$ with respect to y :

$$\begin{aligned} u_{yy} &= (U_\xi)_y + (U_\eta)_y \\ &= (U_{\xi\xi} + U_{\xi\eta}) + (U_{\eta\xi} + U_{\eta\eta}) \\ &= U_{\xi\xi} + 2U_{\xi\eta} + U_{\eta\eta}. \end{aligned}$$

Substitute And Watch The Pure Derivatives Cancel

The original principal expression is:

$$u_{xx} - 4u_{xy} + 3u_{yy}.$$

Substitute the transformed derivatives:

$$\begin{aligned} u_{xx} - 4u_{xy} + 3u_{yy} &= (U_{\xi\xi} + 6U_{\xi\eta} + 9U_{\eta\eta}) \\ &\quad - 4(U_{\xi\xi} + 4U_{\xi\eta} + 3U_{\eta\eta}) \\ &\quad + 3(U_{\xi\xi} + 2U_{\xi\eta} + U_{\eta\eta}). \end{aligned}$$

Collect the coefficient of each new second derivative:

$$\begin{aligned} u_{xx} - 4u_{xy} + 3u_{yy} &= (1 - 4 + 3)U_{\xi\xi} \\ &\quad + (6 - 16 + 6)U_{\xi\eta} \\ &\quad + (9 - 12 + 3)U_{\eta\eta}. \end{aligned}$$

Simplify the three numerical coefficients:

$$u_{xx} - 4u_{xy} + 3u_{yy} = -4U_{\xi\eta}.$$

Therefore the PDE becomes:

$$-4U_{\xi\eta} = 0.$$

Divide by -4 :

$$\boxed{U_{\xi\eta} = 0.}$$

This is the hyperbolic canonical form.

Solve The Canonical Equation

Start from:

$$U_{\xi\eta} = 0.$$

Treat U_{ξ} as the quantity being differentiated with respect to η :

$$\frac{\partial}{\partial \eta}(U_{\xi}) = 0.$$

Therefore U_{ξ} does not depend on η . Write:

$$U_{\xi} = F'(\xi),$$

where F is an arbitrary differentiable function.

Integrate with respect to ξ :

$$U(\xi, \eta) = F(\xi) + G(\eta),$$

where G is the arbitrary function introduced by the integration.

Return to the original coordinates:

$$\xi = x + y, \quad \eta = 3x + y.$$

Hence:

$$\boxed{u(x, y) = F(x + y) + G(3x + y)}.$$

The solution contains one arbitrary function associated with each characteristic family.

Verify The General Family

Let:

$$u(x, y) = F(x + y) + G(3x + y).$$

Differentiate once with respect to x :

$$u_x = F'(x + y) + 3G'(3x + y).$$

Differentiate again with respect to x :

$$u_{xx} = F''(x + y) + 9G''(3x + y).$$

Differentiate u_x with respect to y :

$$u_{xy} = F''(x + y) + 3G''(3x + y).$$

Differentiate twice with respect to y :

$$u_{yy} = F''(x + y) + G''(3x + y).$$

Substitute these derivatives into the original left side:

$$\begin{aligned} u_{xx} - 4u_{xy} + 3u_{yy} &= [F'' + 9G''] - 4[F'' + 3G''] + 3[F'' + G''] \\ &= (1 - 4 + 3)F'' + (9 - 12 + 3)G'' \\ &= 0. \end{aligned}$$

Thus every twice-differentiable pair F, G produces a solution.

Block 4 Summary

For a hyperbolic equation, the two characteristic families provide two new coordinates. In this example they reduced the principal part to $U_{\xi\eta}$, and integrating the canonical equation produced one arbitrary function of each characteristic coordinate.

A Parabolic Equation In Canonical Coordinates

The Goal Of This Block

The goal is to see what changes when the characteristic equation has one repeated direction rather than two distinct directions.

Classify The Equation

Consider:

$$u_{xx} + 2u_{xy} + u_{yy} = 0.$$

The principal coefficients are:

$$A = 1, \quad B = 1, \quad C = 1.$$

Calculate:

$$\begin{aligned} \Delta &= B^2 - AC \\ &= 1^2 - (1)(1) \\ &= 0. \end{aligned}$$

Therefore the equation is parabolic.

Find The Repeated Characteristic

Use the slope equation:

$$Am^2 - 2Bm + C = 0.$$

Substitute $A = 1$, $B = 1$, and $C = 1$:

$$m^2 - 2m + 1 = 0.$$

Factor:

$$(m - 1)^2 = 0.$$

The repeated slope is:

$$m = 1.$$

Thus:

$$\frac{dy}{dx} = 1.$$

Integrate:

$$y = x + k.$$

Rearrange:

$$y - x = k.$$

Since $y - x = k$, choose the characteristic coordinate:

$$\tilde{\zeta} = y - x.$$

A second coordinate must be independent of $\tilde{\zeta}$. A convenient choice is:

$$\eta = x.$$

Check That The Coordinates Are Independent

For:

$$\tilde{\zeta} = y - x, \quad \eta = x,$$

the first derivatives are:

$$\begin{aligned} \tilde{\zeta}_x &= -1, & \tilde{\zeta}_y &= 1, \\ \eta_x &= 1, & \eta_y &= 0. \end{aligned}$$

The Jacobian is:

$$\begin{aligned} J &= \tilde{\zeta}_x \eta_y - \tilde{\zeta}_y \eta_x \\ &= (-1)(0) - (1)(1) \\ &= -1. \end{aligned}$$

Since $J \neq 0$, the coordinate change is valid.

Transform Each Second Derivative

Write $u(x, y) = U(\xi, \eta)$. First:

$$\begin{aligned} u_x &= U_\xi \xi_x + U_\eta \eta_x \\ &= -U_\xi + U_\eta. \end{aligned}$$

Differentiate with respect to x :

$$\begin{aligned} u_{xx} &= -(U_\xi)_x + (U_\eta)_x \\ &= -(-U_{\xi\xi} + U_{\xi\eta}) + (-U_{\eta\xi} + U_{\eta\eta}) \\ &= U_{\xi\xi} - 2U_{\xi\eta} + U_{\eta\eta}. \end{aligned}$$

Next:

$$\begin{aligned} u_y &= U_\xi \xi_y + U_\eta \eta_y \\ &= U_\xi. \end{aligned}$$

Differentiate $u_x = -U_\xi + U_\eta$ with respect to y :

$$\begin{aligned} u_{xy} &= -(U_\xi)_y + (U_\eta)_y \\ &= -U_{\xi\xi} + U_{\eta\xi} \\ &= -U_{\xi\xi} + U_{\xi\eta}. \end{aligned}$$

Differentiate $u_y = U_\xi$ with respect to y :

$$u_{yy} = U_{\xi\xi}.$$

Reduce And Solve The Canonical Equation

Substitute the transformed derivatives into:

$$u_{xx} + 2u_{xy} + u_{yy}.$$

This gives:

$$\begin{aligned} u_{xx} + 2u_{xy} + u_{yy} &= (U_{\xi\xi} - 2U_{\xi\eta} + U_{\eta\eta}) \\ &\quad + 2(-U_{\xi\xi} + U_{\xi\eta}) + U_{\xi\xi}. \end{aligned}$$

Collect like derivatives:

$$\begin{aligned} u_{xx} + 2u_{xy} + u_{yy} &= (1 - 2 + 1)U_{\xi\xi} \\ &\quad + (-2 + 2)U_{\xi\eta} \\ &\quad + U_{\eta\eta} \\ &= U_{\eta\eta}. \end{aligned}$$

Therefore the canonical equation is:

$$\boxed{U_{\eta\eta} = 0.}$$

Integrate once with respect to η :

$$U_{\eta} = F(\xi).$$

Integrate a second time with respect to η :

$$U(\xi, \eta) = \eta F(\xi) + G(\xi).$$

Return to $\xi = y - x$ and $\eta = x$:

$$u(x, y) = xF(y - x) + G(y - x).$$

Both arbitrary functions depend on the single repeated characteristic coordinate.

Verify The Parabolic Solution Family

Let:

$$s = y - x$$

and write:

$$u(x, y) = xF(s) + G(s).$$

Since:

$$s_x = -1, \quad s_y = 1,$$

differentiate with respect to x :

$$\begin{aligned} u_x &= F(s) + xF'(s)s_x + G'(s)s_x \\ &= F(s) - xF'(s) - G'(s). \end{aligned}$$

Differentiate this expression with respect to x :

$$\begin{aligned} u_{xx} &= F'(s)s_x - [F'(s) + xF''(s)s_x] - G''(s)s_x \\ &= -F'(s) - F'(s) + xF''(s) + G''(s) \\ &= -2F'(s) + xF''(s) + G''(s). \end{aligned}$$

Differentiate u_x with respect to y :

$$\begin{aligned} u_{xy} &= F'(s)s_y - xF''(s)s_y - G''(s)s_y \\ &= F'(s) - xF''(s) - G''(s). \end{aligned}$$

Differentiate u with respect to y :

$$u_y = xF'(s) + G'(s).$$

Differentiate again with respect to y :

$$u_{yy} = xF''(s) + G''(s).$$

Now substitute:

$$\begin{aligned} u_{xx} + 2u_{xy} + u_{yy} &= [-2F' + xF'' + G''] \\ &\quad + 2[F' - xF'' - G''] \\ &\quad + [xF'' + G''] \\ &= 0. \end{aligned}$$

The family satisfies the original PDE.

Block 5 Summary

A parabolic equation has one repeated characteristic family. One new coordinate follows that family; the other is any convenient independent coordinate. The canonical principal part contains one pure second derivative.

An Elliptic Equation And Rotation Of Axes

The Goal Of This Block

The goal is to remove a mixed derivative from an elliptic equation even though there are no real characteristic curves.

Why Rotation Is The Natural Tool

Consider a constant-coefficient principal part:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy}.$$

Use a rotation:

$$\begin{aligned}\xi &= x \cos \theta + y \sin \theta, \\ \eta &= -x \sin \theta + y \cos \theta.\end{aligned}$$

The Jacobian is:

$$\begin{aligned}J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (\cos \theta)(\cos \theta) - (\sin \theta)(-\sin \theta) \\ &= \cos^2 \theta + \sin^2 \theta \\ &= 1.\end{aligned}$$

Thus every such rotation is invertible.

After substitution, the coefficient of $U_{\xi\eta}$ is:

$$(C - A) \sin(2\theta) + 2B \cos(2\theta).$$

Set this coefficient equal to zero:

$$(C - A) \sin(2\theta) + 2B \cos(2\theta) = 0.$$

When the relevant denominator is nonzero, rearrange to obtain:

$$\tan(2\theta) = \frac{2B}{A - C}.$$

This angle aligns the new axes with the principal directions of the quadratic form.

A Worked Elliptic Rotation

Consider:

$$2u_{xx} + 2u_{xy} + 2u_{yy} = 0.$$

The coefficients are:

$$A = 2, \quad B = 1, \quad C = 2.$$

Calculate:

$$\begin{aligned} \Delta &= B^2 - AC \\ &= 1 - (2)(2) \\ &= -3 < 0. \end{aligned}$$

The equation is elliptic.

Since $A = C$ and $B \neq 0$, the rotation condition is satisfied by:

$$2\theta = \frac{\pi}{2},$$

so choose:

$$\theta = \frac{\pi}{4}.$$

The rotated coordinates are:

$$\boxed{\xi = \frac{x+y}{\sqrt{2}}, \quad \eta = \frac{-x+y}{\sqrt{2}}}.$$

Transform The Rotated Derivatives

For the chosen rotation:

$$\begin{aligned} \xi_x &= \frac{1}{\sqrt{2}}, & \xi_y &= \frac{1}{\sqrt{2}}, \\ \eta_x &= -\frac{1}{\sqrt{2}}, & \eta_y &= \frac{1}{\sqrt{2}}. \end{aligned}$$

The first derivatives are:

$$\begin{aligned} u_x &= \frac{1}{\sqrt{2}}U_\xi - \frac{1}{\sqrt{2}}U_\eta, \\ u_y &= \frac{1}{\sqrt{2}}U_\xi + \frac{1}{\sqrt{2}}U_\eta. \end{aligned}$$

Differentiate u_x with respect to x :

$$u_{xx} = \frac{1}{2}U_{\xi\xi} - U_{\xi\eta} + \frac{1}{2}U_{\eta\eta}.$$

Differentiate u_x with respect to y :

$$u_{xy} = \frac{1}{2}U_{\xi\xi} - \frac{1}{2}U_{\eta\eta}.$$

Differentiate u_y with respect to y :

$$u_{yy} = \frac{1}{2}U_{\xi\xi} + U_{\xi\eta} + \frac{1}{2}U_{\eta\eta}.$$

Substitute Into The Elliptic Equation

The original left side is:

$$2u_{xx} + 2u_{xy} + 2u_{yy}.$$

Substitute the rotated derivatives:

$$2u_{xx} = U_{\xi\xi} - 2U_{\xi\eta} + U_{\eta\eta},$$

$$2u_{xy} = U_{\xi\xi} - U_{\eta\eta},$$

$$2u_{yy} = U_{\xi\xi} + 2U_{\xi\eta} + U_{\eta\eta}.$$

Add the three lines:

$$\begin{aligned} 2u_{xx} + 2u_{xy} + 2u_{yy} &= (1 + 1 + 1)U_{\xi\xi} \\ &\quad + (-2 + 0 + 2)U_{\xi\eta} \\ &\quad + (1 - 1 + 1)U_{\eta\eta} \\ &= 3U_{\xi\xi} + U_{\eta\eta}. \end{aligned}$$

Thus $2u_{xx} + 2u_{xy} + 2u_{yy} = 0$ becomes:

$$\boxed{3U_{\xi\xi} + U_{\eta\eta} = 0}.$$

The mixed derivative has disappeared, and both remaining pure derivatives have coefficients with the same sign.

Rescale To The Laplace Form

Start from:

$$3U_{\xi\xi} + U_{\eta\eta} = 0.$$

Define:

$$X = \frac{\xi}{\sqrt{3}}, \quad Y = \eta,$$

and write:

$$U(\xi, \eta) = V(X, Y).$$

Since $X_\xi = 1/\sqrt{3}$:

$$\begin{aligned} U_\xi &= V_X X_\xi \\ &= \frac{1}{\sqrt{3}} V_X. \end{aligned}$$

Differentiate once more with respect to ξ :

$$\begin{aligned} U_{\xi\xi} &= \frac{1}{\sqrt{3}} (V_X)_\xi \\ &= \frac{1}{\sqrt{3}} V_{XX} X_\xi \\ &= \frac{1}{3} V_{XX}. \end{aligned}$$

Because $Y = \eta$:

$$U_{\eta\eta} = V_{YY}.$$

Substitute both derivatives:

$$3 \left(\frac{1}{3} V_{XX} \right) + V_{YY} = 0.$$

Simplify:

$$\boxed{V_{XX} + V_{YY} = 0}.$$

Thus the equation has been reduced to Laplace's equation in scaled, rotated coordinates.

What Elliptic Canonical Form Does Not Give Us

For the hyperbolic and parabolic examples, the canonical PDE integrated directly to an arbitrary-function family. Laplace's equation does not collapse in the same way.

Solving an elliptic problem normally requires information on a surrounding boundary. The coordinate transformation simplifies the differential operator, but it does not replace the boundary-value problem.

canonical form reveals structure; it is not automatically a complete solution method.

Block 6 Summary

An elliptic equation has no real characteristic directions. A rotation removes the mixed derivative, and a positive rescaling can reduce the principal part to a sum of two pure second derivatives.

Variable Coefficients And A Change Of Type

The Goal Of This Block

The goal is to classify a variable-coefficient equation by region and to find its real characteristic curves only where they actually exist.

A Mixed-Type Model

Consider:

$$y u_{xx} + u_{yy} = 0.$$

The coefficients are:

$$A = y, \quad B = 0, \quad C = 1.$$

Hence:

$$\begin{aligned} \Delta &= B^2 - AC \\ &= 0 - y(1) \\ &= -y. \end{aligned}$$

Therefore:

$$\begin{array}{l|l} y < 0 & \Delta > 0: \text{hyperbolic} \\ y = 0 & \Delta = 0: \text{parabolic} \\ y > 0 & \Delta < 0: \text{elliptic} \end{array}$$

The line $y = 0$ separates regions with different PDE behaviour.

Find Characteristics In The Hyperbolic Region

Use the differential characteristic equation:

$$A(dy)^2 - 2B dx dy + C(dx)^2 = 0.$$

Insert $A = y$, $B = 0$, and $C = 1$:

$$y(dy)^2 + (dx)^2 = 0.$$

In the hyperbolic region $y < 0$, rearrange:

$$(dx)^2 = -y(dy)^2.$$

Take square roots:

$$\frac{dx}{dy} = \pm \sqrt{-y}.$$

Using dx/dy avoids dividing by $A = y$ and gives a cleaner integral near the change-of-type line.

Integrate The Two Variable-Coefficient Families

For the positive sign:

$$\frac{dx}{dy} = \sqrt{-y}.$$

Integrate with respect to y :

$$x = \int \sqrt{-y} dy + k_1.$$

Let $q = -y$, so $dq = -dy$. Then:

$$\begin{aligned}\int \sqrt{-y} dy &= - \int q^{1/2} dq \\ &= -\frac{2}{3} q^{3/2} \\ &= -\frac{2}{3} (-y)^{3/2}.\end{aligned}$$

Thus:

$$x + \frac{2}{3}(-y)^{3/2} = k_1.$$

For the negative sign:

$$\frac{dx}{dy} = -\sqrt{-y}.$$

Integrating gives:

$$x - \frac{2}{3}(-y)^{3/2} = k_2.$$

Therefore characteristic coordinates in $y < 0$ may be chosen as:

$$\boxed{\zeta = x + \frac{2}{3}(-y)^{3/2}, \quad \eta = x - \frac{2}{3}(-y)^{3/2}}.$$

Check The Variable-Coefficient Jacobian

Differentiate the coordinates for $y < 0$:

$$\begin{aligned}\zeta_x &= 1, & \zeta_y &= -\sqrt{-y}, \\ \eta_x &= 1, & \eta_y &= \sqrt{-y}.\end{aligned}$$

For $\xi = x + \frac{2}{3}(-y)^{3/2}$ and $\eta = x - \frac{2}{3}(-y)^{3/2}$, calculate the Jacobian:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (1) \sqrt{-y} - (-\sqrt{-y})(1) \\ &= 2\sqrt{-y}. \end{aligned}$$

For $y < 0$, $\sqrt{-y} > 0$, so $J \neq 0$. As y approaches 0 from below, J approaches 0.

This degeneration is not an algebra accident. It records the fact that the two hyperbolic characteristic directions merge at the parabolic boundary.

Why Extra First-Derivative Terms Appear

The coordinates ξ and η contain $(-y)^{3/2}$, so they are not linear. Consequently:

$$\xi_{yy} \neq 0, \quad \eta_{yy} \neq 0.$$

The second-derivative chain rule contains:

$$U_\xi \xi_{yy} + U_\eta \eta_{yy}.$$

These terms survive as first-order derivatives of U in the transformed PDE. The characteristic choice still simplifies the principal part, but it does not guarantee that every lower-order term disappears.

Block 7 Summary

For variable coefficients, classify point by point or region by region. Characteristics exist as real curves only in hyperbolic or parabolic regions. Nonlinear coordinate changes generally create lower-order chain-rule terms, and the Jacobian may reveal where the transformation degenerates.

Dimensional Analysis And Nondimensionalisation

The Goal Of This Block

The goal is to use physical dimensions as an error check and then rescale a PDE so that its essential dimensionless parameters become visible.

Dimensions Are Not The Same As Units

A unit might be metres or centimetres. Both represent the dimension of length. We use:

$$[x] = L, \quad [t] = T,$$

where L denotes length and T denotes time.

If u represents temperature, we may write:

$$[u] = \Theta.$$

Every term added in a physically meaningful equation must have the same dimension.²

Find The Dimension Of Thermal Diffusivity

Consider the heat equation:

$$u_t = \kappa u_{xx}.$$

² Dimensions describe the kind of quantity, while units specify a measurement scale. Changing metres to centimetres changes a numerical value but does not change the dimension L .

For the heat equation $u_t = \kappa u_{xx}$, differentiation with respect to time divides by time:

$$[u_t] = \frac{\Theta}{T}.$$

Two derivatives with respect to space divide by length twice:

$$[u_{xx}] = \frac{\Theta}{L^2}.$$

The two sides must have the same dimension:

$$\frac{\Theta}{T} = [\kappa] \frac{\Theta}{L^2}.$$

Cancel Θ :

$$\frac{1}{T} = \frac{[\kappa]}{L^2}.$$

Multiply by L^2 :

$$\boxed{[\kappa] = \frac{L^2}{T}}.$$

Thermal diffusivity therefore has dimensions of area per unit time.

Check The Wave-Speed Dimension

For the wave equation:

$$u_{tt} = c^2 u_{xx},$$

For $u_{tt} = c^2 u_{xx}$, the derivative dimensions are:

$$[u_{tt}] = \frac{[u]}{T^2}, \quad [u_{xx}] = \frac{[u]}{L^2}.$$

Match the dimensions of both sides:

$$\frac{[u]}{T^2} = [c]^2 \frac{[u]}{L^2}.$$

Cancel $[u]$:

$$\frac{1}{T^2} = \frac{[c]^2}{L^2}.$$

Multiply by L^2 :

$$[c]^2 = \frac{L^2}{T^2}.$$

Take the positive square root for a speed:

$$\boxed{[c] = \frac{L}{T}}.$$

Detect A Dimensionally Inconsistent Model

Suppose someone proposes:

$$u_t = u_{xx} + u$$

for a dimensional time t and length x .

For the proposed model $u_t = u_{xx} + u$, the three terms have dimensions:

$$[u_t] = \frac{[u]}{T}, \quad [u_{xx}] = \frac{[u]}{L^2}, \quad [u] = [u].$$

These dimensions do not match. Coefficients with appropriate dimensions are missing.

A consistent reaction-diffusion model may instead be:

$$u_t = Du_{xx} + ru,$$

where:

$$[D] = \frac{L^2}{T}, \quad [r] = \frac{1}{T}.$$

Then:

$$[Du_{xx}] = \frac{L^2}{T} \frac{[u]}{L^2} = \frac{[u]}{T},$$

and:

$$[ru] = \frac{1}{T} [u] = \frac{[u]}{T}.$$

All three terms now agree.

Nondimensionalise The Heat Equation

Consider:

$$u_t = \kappa u_{xx}$$

on a spatial interval with characteristic length L . Let U_0 be a typical size of u .

Define dimensionless variables:

$$X = \frac{x}{L}, \quad \tau = \frac{\kappa t}{L^2}, \quad U(X, \tau) = \frac{u(x, t)}{U_0}.$$

Equivalently:

$$x = LX, \quad t = \frac{L^2}{\kappa} \tau, \quad u = U_0 U.$$

The time scale:

$$t_c = \frac{L^2}{\kappa}$$

is the diffusion time across distance L .

Transform The Heat Derivatives One At A Time

Since:

$$\tau = \frac{\kappa t}{L^2},$$

we have:

$$\frac{\partial \tau}{\partial t} = \frac{\kappa}{L^2}.$$

Differentiate $u = U_0 U(X, \tau)$ with respect to t . The coordinate $X = x/L$ does not depend on t , so:

$$\begin{aligned} u_t &= U_0 U_\tau \frac{\partial \tau}{\partial t} \\ &= U_0 \frac{\kappa}{L^2} U_\tau. \end{aligned}$$

For the spatial derivative:

$$\frac{\partial X}{\partial x} = \frac{1}{L}.$$

Therefore:

$$\begin{aligned} u_x &= U_0 U_X \frac{\partial X}{\partial x} \\ &= \frac{U_0}{L} U_X. \end{aligned}$$

Differentiate once more with respect to x :

$$\begin{aligned} u_{xx} &= \frac{U_0}{L} (U_X)_x \\ &= \frac{U_0}{L} U_{XX} \frac{\partial X}{\partial x} \\ &= \frac{U_0}{L^2} U_{XX}. \end{aligned}$$

Cancel The Dimensional Scales

Substitute:

$$u_t = U_0 \frac{\kappa}{L^2} U_\tau$$

and:

$$u_{xx} = \frac{U_0}{L^2} U_{XX}$$

Substitute these transformed derivatives into $u_t = \kappa u_{xx}$:

$$U_0 \frac{\kappa}{L^2} U_\tau = \kappa \left(\frac{U_0}{L^2} U_{XX} \right).$$

The common factor is:

$$\frac{\kappa U_0}{L^2}.$$

Divide both sides by this nonzero factor:

$$\boxed{U_\tau = U_{XX}}.$$

The dimensionless heat equation contains no material parameter. The diffusivity has been absorbed into the time scale.

Advection Versus Diffusion

Consider:

$$u_t + v u_x = D u_{xx},$$

where v is a constant transport speed and D is a diffusivity.

Use the transport time L/v :

$$X = \frac{x}{L}, \quad \tau = \frac{vt}{L}, \quad u = U_0 U.$$

The derivatives become:

$$u_t = \frac{U_0 v}{L} U_\tau, \quad u_x = \frac{U_0}{L} U_X, \quad u_{xx} = \frac{U_0}{L^2} U_{XX}.$$

Substitute these derivatives into $u_t + vu_x = Du_{xx}$:

$$\frac{U_0 v}{L} U_\tau + v \frac{U_0}{L} U_X = D \frac{U_0}{L^2} U_{XX}.$$

Divide by $U_0 v/L$:

$$U_\tau + U_X = \frac{D}{vL} U_{XX}.$$

Define the Peclet number:

$$\boxed{\text{Pe} = \frac{vL}{D}}.$$

Since $D/(vL) = 1/\text{Pe}$, the dimensionless equation is:

$$\boxed{U_\tau + U_X = \frac{1}{\text{Pe}} U_{XX}}.$$

A large Peclet number means transport by advection is strong relative to diffusion on the chosen length scale.

Check That The Peclet Number Is Dimensionless

Use:

$$[v] = \frac{L}{T}, \quad [L] = L, \quad [D] = \frac{L^2}{T}.$$

Then:

$$\begin{aligned} [\text{Pe}] &= \frac{[v][L]}{[D]} \\ &= \frac{(L/T)(L)}{L^2/T} \\ &= \frac{L^2/T}{L^2/T} \\ &= 1. \end{aligned}$$

The symbol 1 means the group has no physical dimension.

Block 8 Summary

Dimensional consistency checks whether terms can legitimately be added. Nondimensionalisation goes further: it chooses natural scales, transforms each derivative, and exposes the parameter combinations that control the behaviour.

A Reliable Classification And Transformation Workflow

The Goal Of This Block

The goal is to collect the chapter into a sequence that can be applied without guessing which formula comes next.

The Full Workflow

Step 1: Isolate the principal part. Write it in the convention:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy}.$$

Step 2: Read A , B , and C . If the visible mixed coefficient is q , solve $2B = q$ before continuing.

Step 3: Calculate the discriminant.

$$\Delta = B^2 - AC.$$

Step 4: Classify by region. Variable coefficients may make the sign of Δ depend on (x, y) .

Step 5: For hyperbolic or parabolic equations, write the characteristic equation.

$$A(dy)^2 - 2B dx dy + C(dx)^2 = 0.$$

Step 6: Solve for characteristic directions. Use dy/dx or dx/dy , whichever avoids a zero denominator or an awkward integral.

Step 7: Integrate each direction field. Convert the constants of integration into coordinate functions.

Step 8: Choose enough independent coordinates. Hyperbolic equations use the two characteristic families. Parabolic equations need the repeated characteristic coordinate plus another independent coordinate.

Step 9: Check the Jacobian.

$$J = \xi_x \eta_y - \xi_y \eta_x \neq 0.$$

Step 10: Transform derivatives locally. Write the first derivatives, then the second derivatives, and only then substitute into the PDE.

Step 11: Simplify the canonical equation. Do not assume lower-order terms vanish when the coordinate functions are nonlinear.

Step 12: Transform boundary or initial data. A new PDE without transformed data is not yet the complete transformed problem.

Step 13: Verify the result. If a general family is obtained, differentiate it and substitute into the original equation.

Common Mistake: Using The Visible Mixed Coefficient As B

For:

$$u_{xx} + 6u_{xy} + 5u_{yy} = 0,$$

the convention is:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} = 0.$$

Therefore:

$$A = 1, \quad 2B = 6, \quad C = 5,$$

Solving $2B = 6$ gives:

$$B = 3.$$

The correct discriminant is:

$$\Delta = 3^2 - (1)(5) = 4.$$

Using $B = 6$ would produce the wrong discriminant and the wrong characteristic slopes.

Common Mistake: Calling A Slope A Coordinate

Suppose the slope equation gives:

$$\frac{dy}{dx} = 2.$$

The number 2 is a direction, not a coordinate. Integrate:

$$y = 2x + k.$$

Then rearrange:

$$y - 2x = k.$$

The coordinate that stays constant along this family is:

$$\zeta = y - 2x.$$

Common Mistake: Skipping The Jacobian

Consider:

$$\xi = x + y, \quad \eta = 2x + 2y.$$

Although two formulas are written, the second is only twice the first:

$$\eta = 2\xi.$$

Calculate:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (1)(2) - (1)(2) \\ &= 0. \end{aligned}$$

The proposed transformation collapses the plane onto one coordinate direction and cannot be inverted.

Common Mistake: Dropping Lower-Order Chain-Rule Terms

For nonlinear $\xi(x, y)$:

$$u_{xx} = U_{\xi\xi} \xi_x^2 + \cdots + U_{\xi} \xi_{xx} + \cdots.$$

The term $U_{\xi} \xi_{xx}$ vanishes only when $\xi_{xx} = 0$. A characteristic choice removes a selected principal coefficient; it does not make every second derivative of the coordinate functions zero.

Common Mistake: Treating Canonical Form As The Final Answer

Reducing a PDE to:

$$U_{\xi\eta} = 0$$

is an important structural step. If the question asks for a solution, continue to:

$$U(\xi, \eta) = F(\xi) + G(\eta)$$

and then return to x, y .

If the original problem includes boundary data, rewrite those curves and data in the new coordinates as well.

Block 9 Summary

The dependable chain is:

principal part → **coefficients** → **discriminant** → **characteristic directions** → **characteristic curves** → **Jacobian** → **transformed PDE** → **transformed data.**

Each arrow represents a calculation that should be shown rather than implied.

Original Practice Set

Problems 1 To 4: Classification And Characteristics

Problem 1: Read The Principal Coefficients. Classify:

$$4u_{xx} - 12u_{xy} + 5u_{yy} + 7u_x - u = 0.$$

Problem 2: Classify By Region. Classify:

$$(x - 1)u_{xx} + u_{yy} = 0$$

in the regions $x < 1$, $x = 1$, and $x > 1$.

Problem 3: Find Two Characteristic Slopes. Find the characteristic slopes of:

$$u_{xx} - u_{xy} - 2u_{yy} = 0.$$

Problem 4: Build Characteristic Coordinates. For the equation in Problem 3, integrate both characteristic directions, choose ξ and η , and verify that the Jacobian is nonzero.

Problems 5 To 8: Canonical Forms

Problem 5: Solve A Hyperbolic Equation. Reduce and solve:

$$u_{xx} + 3u_{xy} + 2u_{yy} = 0.$$

Problem 6: Solve A Parabolic Equation. Reduce and solve:

$$4u_{xx} + 4u_{xy} + u_{yy} = 0.$$

Problem 7: Rotate An Elliptic Equation. Use:

$$\xi = \frac{x+y}{\sqrt{2}}, \quad \eta = \frac{-x+y}{\sqrt{2}}$$

to remove the mixed derivative from:

$$3u_{xx} + 2u_{xy} + 3u_{yy} = 0.$$

Problem 8: Check A Proposed Transformation. Determine whether:

$$\xi = x - 2y, \quad \eta = 3x + y$$

defines a valid coordinate change.

Problems 9 To 12: Dimensions And Scaling

Problem 9: Find A Coefficient Dimension. For:

$$u_t = \alpha u_{xxxx},$$

find the dimension of α .

Problem 10: Scale A Heat Equation. For:

$$u_t = 5u_{xx}, \quad 0 < x < 4,$$

choose dimensionless X and τ that reduce the PDE to $U_\tau = U_{XX}$.

Problem 11: Calculate A Peclet Number. An advection-diffusion model has:

$$v = 0.4 \text{ m s}^{-1}, \quad L = 2 \text{ m}, \quad D = 0.08 \text{ m}^2 \text{ s}^{-1}.$$

Calculate $\text{Pe} = vL/D$ and interpret its size.

Problem 12: Check Dimensional Consistency. Suppose $[u] = \Theta$, $[x] = L$, and $[t] = T$. Determine the dimensions that a and b need in:

$$u_t = au_{xx} + bu.$$

Problems 13 And 14: Integration And Diagnosis

Problem 13: Integrate A Forced Canonical Equation. Find the general solution of:

$$U_{\xi\eta} = e^{\xi}.$$

Problem 14: Diagnose A Failed Coordinate Pair. Explain precisely why:

$$\xi = x + y, \quad \eta = 2x + 2y$$

cannot be used as two new coordinates.

Worked Answers: Problems 1 To 4

Answer 1. The equation is:

$$4u_{xx} - 12u_{xy} + 5u_{yy} + 7u_x - u = 0.$$

Match its principal part with:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy}.$$

Therefore:

$$A = 4, \quad 2B = -12, \quad C = 5.$$

Solve for B :

$$B = -6.$$

Now calculate:

$$\begin{aligned}\Delta &= B^2 - AC \\ &= (-6)^2 - (4)(5) \\ &= 36 - 20 \\ &= 16.\end{aligned}$$

Since $\Delta > 0$:

the equation is hyperbolic.

The terms $7u_x$ and $-u$ are lower order and do not enter the discriminant.

Answer 2. For:

$$(x - 1)u_{xx} + u_{yy} = 0,$$

the principal coefficients are:

$$A = x - 1, \quad B = 0, \quad C = 1.$$

Therefore:

$$\begin{aligned} \Delta &= B^2 - AC \\ &= 0 - (x - 1)(1) \\ &= 1 - x. \end{aligned}$$

If $x < 1$, then $1 - x > 0$, so the equation is hyperbolic.

If $x = 1$, then $1 - x = 0$, so the equation is parabolic.

If $x > 1$, then $1 - x < 0$, so the equation is elliptic.

Thus:

$x < 1$	hyperbolic
$x = 1$	parabolic
$x > 1$	elliptic

Answer 3. For:

$$u_{xx} - u_{xy} - 2u_{yy} = 0,$$

Compare $u_{xx} - u_{xy} - 2u_{yy} = 0$ with:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} = 0.$$

This gives:

$$A = 1, \quad 2B = -1, \quad C = -2,$$

so:

$$B = -\frac{1}{2}.$$

The characteristic slope equation is:

$$Am^2 - 2Bm + C = 0.$$

Substitute $A = 1$, $B = -1/2$, and $C = -2$:

$$m^2 - 2\left(-\frac{1}{2}\right)m - 2 = 0.$$

Simplify the middle coefficient:

$$m^2 + m - 2 = 0.$$

Factor:

$$(m - 1)(m + 2) = 0.$$

The characteristic slopes are therefore:

$$m_1 = 1, \quad m_2 = -2.$$

Answer 4. The two slopes from Problem 3 are:

$$\frac{dy}{dx} = 1$$

and:

$$\frac{dy}{dx} = -2.$$

Integrate the first direction:

$$y = x + k_1.$$

Rearrange $y = x + k_1$ to isolate the characteristic constant:

$$y - x = k_1.$$

Choose:

$$\xi = y - x.$$

Integrate the second direction:

$$y = -2x + k_2.$$

Rearrange $y = -2x + k_2$:

$$2x + y = k_2.$$

Choose:

$$\eta = 2x + y.$$

The coordinate derivatives are:

$$\begin{aligned}\xi_x &= -1, & \xi_y &= 1, \\ \eta_x &= 2, & \eta_y &= 1.\end{aligned}$$

Calculate the Jacobian:

$$\begin{aligned}J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (-1)(1) - (1)(2) \\ &= -3.\end{aligned}$$

Since $J \neq 0$, the two characteristic coordinates define a valid local coordinate change.

Worked Answers: Problems 5 To 8

Answer 5. Start from:

$$u_{xx} + 3u_{xy} + 2u_{yy} = 0.$$

The principal coefficients in the $2B$ convention are:

$$A = 1, \quad B = \frac{3}{2}, \quad C = 2.$$

The characteristic equation is:

$$Am^2 - 2Bm + C = 0.$$

Substitute the coefficients:

$$m^2 - 3m + 2 = 0.$$

Factor:

$$(m - 1)(m - 2) = 0.$$

Thus the slopes are:

$$m_1 = 1, \quad m_2 = 2.$$

Integrate the first slope:

$$\frac{dy}{dx} = 1 \quad \Rightarrow \quad y - x = k_1.$$

Integrate the second slope:

$$\frac{dy}{dx} = 2 \implies y - 2x = k_2.$$

Choose:

$$\xi = y - x, \quad \eta = y - 2x.$$

The Jacobian is:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (-1)(1) - (1)(-2) \\ &= 1. \end{aligned}$$

The transformation is valid. For a constant-coefficient hyperbolic equation, these two characteristic coordinates reduce the PDE to a nonzero multiple of:

$$U_{\xi\eta} = 0.$$

Integrate the canonical equation:

$$U(\xi, \eta) = F(\xi) + G(\eta).$$

Return to the original variables:

$$\boxed{u(x, y) = F(y - x) + G(y - 2x)}.$$

To check the first component, let $u = F(y - x)$. Then:

$$u_{xx} = F'', \quad u_{xy} = -F'', \quad u_{yy} = F'',$$

so:

$$u_{xx} + 3u_{xy} + 2u_{yy} = F'' - 3F'' + 2F'' = 0.$$

For $u = G(y - 2x)$:

$$u_{xx} = 4G'', \quad u_{xy} = -2G'', \quad u_{yy} = G'',$$

and:

$$u_{xx} + 3u_{xy} + 2u_{yy} = 4G'' - 6G'' + 2G'' = 0.$$

Linearity then verifies their sum.

Answer 6. For:

$$4u_{xx} + 4u_{xy} + u_{yy} = 0,$$

the principal coefficients are:

$$A = 4, \quad B = 2, \quad C = 1.$$

Calculate:

$$\Delta = 2^2 - (4)(1) = 0.$$

For $4u_{xx} + 4u_{xy} + u_{yy} = 0$, $\Delta = 0$, so the equation is parabolic. Its slope equation is:

$$4m^2 - 4m + 1 = 0.$$

Factor:

$$(2m - 1)^2 = 0.$$

The repeated slope is:

$$m = \frac{1}{2}.$$

Integrate:

$$\frac{dy}{dx} = \frac{1}{2} \implies y = \frac{x}{2} + k.$$

Multiply by 2 and rearrange:

$$2y - x = 2k.$$

Choose:

$$\xi = 2y - x.$$

Choose the independent coordinate:

$$\eta = y.$$

For $\xi = 2y - x$ and $\eta = y$, the Jacobian is:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (-1)(1) - (2)(0) \\ &= -1. \end{aligned}$$

Thus the transformation is valid. In these coordinates the operator $4\partial_{xx} + 4\partial_{xy} + \partial_{yy}$ becomes $\partial_{\eta\eta}$, so:

$$U_{\eta\eta} = 0.$$

Integrate twice with respect to η :

$$U(\xi, \eta) = \eta F(\xi) + G(\xi).$$

Return to $\xi = 2y - x$ and $\eta = y$:

$$u(x, y) = yF(2y - x) + G(2y - x).$$

Answer 7. Use:

$$\xi = \frac{x + y}{\sqrt{2}}, \quad \eta = \frac{-x + y}{\sqrt{2}}.$$

The required transformed derivatives are:

$$\begin{aligned} u_{xx} &= \frac{1}{2}U_{\xi\xi} - U_{\xi\eta} + \frac{1}{2}U_{\eta\eta}, \\ u_{xy} &= \frac{1}{2}U_{\xi\xi} - \frac{1}{2}U_{\eta\eta}, \\ u_{yy} &= \frac{1}{2}U_{\xi\xi} + U_{\xi\eta} + \frac{1}{2}U_{\eta\eta}. \end{aligned}$$

Substitute these derivatives into the original principal expression:

$$3u_{xx} + 2u_{xy} + 3u_{yy}.$$

This gives:

$$3u_{xx} = \frac{3}{2}U_{\xi\xi} - 3U_{\xi\eta} + \frac{3}{2}U_{\eta\eta},$$

$$2u_{xy} = U_{\xi\xi} - U_{\eta\eta},$$

$$3u_{yy} = \frac{3}{2}U_{\xi\xi} + 3U_{\xi\eta} + \frac{3}{2}U_{\eta\eta}.$$

Add the three transformed lines:

$$\begin{aligned} 3u_{xx} + 2u_{xy} + 3u_{yy} &= \left(\frac{3}{2} + 1 + \frac{3}{2}\right)U_{\xi\xi} \\ &\quad + (-3 + 3)U_{\xi\eta} \\ &\quad + \left(\frac{3}{2} - 1 + \frac{3}{2}\right)U_{\eta\eta} \\ &= 4U_{\xi\xi} + 2U_{\eta\eta}. \end{aligned}$$

Therefore the canonical equation is:

$$\boxed{2U_{\xi\xi} + U_{\eta\eta} = 0},$$

after division by 2.

Answer 8. For:

$$\xi = x - 2y, \quad \eta = 3x + y,$$

the derivatives are:

$$\xi_x = 1, \quad \xi_y = -2,$$

$$\eta_x = 3, \quad \eta_y = 1.$$

Calculate:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (1)(1) - (-2)(3) \\ &= 1 + 6 \\ &= 7. \end{aligned}$$

Since $J \neq 0$:

the coordinate change is valid.

Worked Answers: Problems 9 To 12**Answer 9.** For:

$$u_t = \alpha u_{xxxx},$$

the time derivative has dimension:

$$[u_t] = \frac{[u]}{T}.$$

Four spatial derivatives give:

$$[u_{xxxx}] = \frac{[u]}{L^4}.$$

Match the dimensions of both sides:

$$\frac{[u]}{T} = [\alpha] \frac{[u]}{L^4}.$$

Cancel $[u]$:

$$\frac{1}{T} = \frac{[\alpha]}{L^4}.$$

Multiply by L^4 :

$$\boxed{[\alpha] = \frac{L^4}{T}}.$$

Answer 10. The heat equation is:

$$u_t = 5u_{xx}$$

on a length-4 interval. Thus:

$$\kappa = 5, \quad L = 4.$$

Use the dimensionless space variable:

$$X = \frac{x}{L}.$$

Substitute $L = 4$:

$$\boxed{X = \frac{x}{4}}.$$

Use the diffusion time variable:

$$\tau = \frac{\kappa t}{L^2}.$$

Insert $\kappa = 5$ and $L = 4$:

$$\begin{aligned} \tau &= \frac{5t}{4^2} \\ &= \frac{5t}{16}. \end{aligned}$$

The numerical calculation therefore gives:

$$\boxed{\tau = \frac{5t}{16}}.$$

With $u = U_0 U(X, \tau)$, the derivative factors are:

$$u_t = U_0 \frac{5}{16} U_\tau$$

and:

$$u_{xx} = U_0 \frac{1}{16} U_{XX}.$$

Substitute into $u_t = 5u_{xx}$:

$$U_0 \frac{5}{16} U_\tau = 5 \left(U_0 \frac{1}{16} U_{XX} \right).$$

Divide by $5U_0/16$:

$$\boxed{U_\tau = U_{XX}}.$$

Answer 11. Use:

$$\text{Pe} = \frac{vL}{D}.$$

Substitute:

$$v = 0.4 \text{ m s}^{-1}, \quad L = 2 \text{ m}, \quad D = 0.08 \text{ m}^2 \text{ s}^{-1}.$$

Then:

$$\begin{aligned} \text{Pe} &= \frac{(0.4)(2)}{0.08} \\ &= \frac{0.8}{0.08} \\ &= 10. \end{aligned}$$

The Peclet-number calculation gives:

$$\boxed{\text{Pe} = 10.}$$

On this length scale, advection is stronger than diffusion, although diffusion is not absent.

Answer 12. For:

$$u_t = au_{xx} + bu,$$

the left side has dimension:

$$[u_t] = \frac{\Theta}{T}.$$

Since:

$$[u_{xx}] = \frac{\Theta}{L^2},$$

the diffusion term must satisfy:

$$[a] \frac{\Theta}{L^2} = \frac{\Theta}{T}.$$

Cancel Θ and solve for $[a]$:

$$\boxed{[a] = \frac{L^2}{T}}.$$

For the reaction term bu in the same equation:

$$[b]\Theta = \frac{\Theta}{T}.$$

Cancel Θ :

$$\boxed{[b] = \frac{1}{T}}.$$

Both terms on the right then have the same dimension as u_t .

Worked Answers: Problems 13 And 14

Answer 13. Start from:

$$U_{\xi\eta} = e^{\xi}.$$

Regard $U_{\xi\eta}$ as the η -derivative of U_{ξ} :

$$\frac{\partial}{\partial \eta}(U_{\xi}) = e^{\xi}.$$

Integrate both sides with respect to η . Since e^{ξ} is constant with respect to η :

$$U_{\xi} = \eta e^{\xi} + F(\xi).$$

The integration function is $F(\xi)$ because differentiating it with respect to η gives zero.

Now integrate with respect to ξ :

$$U(\xi, \eta) = \int [\eta e^{\xi} + F(\xi)] d\xi.$$

Integrate each term:

$$U(\xi, \eta) = \eta e^{\xi} + G(\xi) + H(\eta),$$

where:

$$G'(\xi) = F(\xi)$$

and $H(\eta)$ is the arbitrary function introduced by the second integration.

Integrating $U_{\xi\eta} = e^{\xi}$ twice therefore gives:

$$\boxed{U(\xi, \eta) = \eta e^{\xi} + G(\xi) + H(\eta)}.$$

Verify by differentiating with respect to ξ :

$$U_{\xi} = \eta e^{\xi} + G'(\xi).$$

Differentiate this result with respect to η :

$$\begin{aligned} U_{\xi\eta} &= \frac{\partial}{\partial \eta} [\eta e^{\xi} + G'(\xi)] \\ &= e^{\xi} + 0 \\ &= e^{\xi}. \end{aligned}$$

Answer 14. The proposed coordinates are:

$$\xi = x + y, \quad \eta = 2x + 2y.$$

First observe:

$$\eta = 2(x + y).$$

Since $\xi = x + y$, this means:

$$\eta = 2\xi.$$

The two coordinates are therefore not independent.

For $\xi = x + y$ and $\eta = 2x + 2y$, the Jacobian confirms the problem. The derivatives are:

$$\xi_x = 1, \quad \xi_y = 1,$$

$$\eta_x = 2, \quad \eta_y = 2.$$

Hence:

$$\begin{aligned} J &= \xi_x \eta_y - \xi_y \eta_x \\ &= (1)(2) - (1)(2) \\ &= 0. \end{aligned}$$

Because $J = 0$, the map cannot be locally inverted. Specifying ξ already determines η , so the pair provides only one independent direction.

Chapter 32 Final Summary

For a second-order PDE with principal part:

$$Au_{xx} + 2Bu_{xy} + Cu_{yy},$$

the discriminant is:

$$\Delta = B^2 - AC.$$

Its sign gives:

$$\begin{array}{l|l} \Delta > 0 & \text{hyperbolic} \\ \Delta = 0 & \text{parabolic} \\ \Delta < 0 & \text{elliptic} \end{array}$$

A characteristic level curve $\phi(x, y) = \text{constant}$ satisfies:

$$A\phi_x^2 + 2B\phi_x\phi_y + C\phi_y^2 = 0.$$

For a nonvertical curve with $m = dy/dx$, this becomes:

$$Am^2 - 2Bm + C = 0.$$

Hyperbolic equations have two real characteristic directions, parabolic equations have one repeated real direction, and elliptic equations have no real characteristic direction.

A coordinate change:

$$\xi = \xi(x, y), \quad \eta = \eta(x, y)$$

Its Jacobian must satisfy:

$$J = \xi_x \eta_y - \xi_y \eta_x \neq 0.$$

The three typical canonical principal parts are:

$$\begin{aligned} U_{\xi\eta} & \text{ hyperbolic,} \\ U_{\eta\eta} & \text{ parabolic,} \\ U_{\xi\xi} + U_{\eta\eta} & \text{ elliptic after scaling.} \end{aligned}$$

For nonlinear coordinate changes, second derivatives of ξ and η produce lower-order derivatives of U and must not be discarded.

Dimensional analysis requires every additive term to have the same dimension. For the heat and wave equations:

$$[\kappa] = \frac{L^2}{T}, \quad [c] = \frac{L}{T}.$$

Nondimensionalisation transforms both the variables and the derivatives. It can remove dimensional constants or combine them into dimensionless groups such as:

$$\text{Pe} = \frac{vL}{D}.$$

classify the principal part, integrate characteristic directions into coordinate functions, check the Jacobian, transform every required derivative, preserve lower-order terms, and verify the resulting equation.